The Nouns Beneath the Numbers

Ontological Fluency and the Corrigible Objects of Physics

On language, idealisation and the hidden architecture of physical quantities

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School physics presents students with equations whose objects and relations have already been selected, simplified and idealised. Difficulty arises when the algebra is visible while the modelling decisions beneath it remain hidden. Ontological fluency is the ability to recognise what kind of thing each symbol denotes, how it functions within the constructed system, and what observation would require the model to be revised.

Prologue: corrigible objects

Physics begins with phenomena, and with the provisional entities, relations and quantities through which those phenomena become describable. Nature supplies no inventory of objects already labelled, so the labelling is ours. Force, field, charge, energy and particle are conceptual commitments that partition systems, organise observations and make consequences calculable.

Those commitments have changed, and changed radically: Newtonian mechanics organises interaction through force, classical electromagnetism gives the field a local and dynamical role, and quantum field theory treats particles as excitations of fields while representing interactions through structures that need not correspond to directly observable intermediaries. Each framework reorganises what is treated as an object, what is treated as a relation, and what counts as an explanation.

Ontology here means the working inventory a theory asks us to use: its systems, entities, properties, relations, processes and representational constructs, and it makes no claim to a final catalogue of everything that ultimately exists. The objects are provisional, though their provisional status leaves them far from arbitrary.[1] They earn their place through the explanatory and predictive work they perform, and they remain answerable to the measurements, interventions and observations they organise.

Scientific practice joins two dispositions that ordinary language rarely places together. Its desire to know may be incorrigible; its ontology must remain corrigible. We possess a familiar word for what cannot be corrected and scarcely use its opposite. Science depends upon precisely that neglected virtue: corrigibility.[2] Its objects must remain stable enough to reason with and open enough to revision when the world resists them. Force, field and particle are therefore corrigible commitments.

This answerability distinguishes physical language from symbolic production that remains internal to its own rules. Pure mathematics can develop without requiring its objects to enter into contact with physical reality. Physics places its words and equations into that contact. A language model can parse, splice and recite the inherited tokens of such encounters. Considered merely as a language model, it encounters the world through records produced by others rather than through the resistance of an experiment it has undertaken. Scientific practice is organised around that resistance, so that concepts are externalised as claims and the world is permitted to refuse them.

Ontological fluency is the classroom form of this discipline. To ask what kind of thing a symbol denotes is also to ask which system it belongs to, which observations support it, which relations it permits, and what would count as its failure. Before a student can manipulate an equation physically, a world must have been constructed in which its symbols can be wrong.

"Our curiosity may be incorrigible; our ontology must remain corrigible."

School physics is densely populated by nouns. Force, mass, energy, current, field, resistance, acceleration, density, momentum and power enter the classroom as though each named a stable item in the inventory of nature, and the grammar is reassuring: a noun appears to point towards a thing, a definition appears to tell us what that thing is, and an equation appears to state how the named things are related.

Yet these terms do not all denote the same kind of entity. A force is an interaction represented by a vector; acceleration is a rate of change of velocity; density is an intensive ratio property; current is a rate of charge transfer; power is a rate of energy transfer; a field assigns a quantity to locations in space and time; resistance is a relational parameter within a circuit model; energy is a conserved quantity defined relative to a chosen system. To place all of these under the grammatical heading of "things" already hides much of the physics. This essay develops a simple proposition:

> Many persistent errors in school physics are failures to identify what kind of entity, process, rela-

[1] The position is a fallibilist one, in Peirce's sense and later Popper's: any claim may turn out to be mistaken, and inquiry is rational precisely because it is organised to find out. Fallibilism should be distinguished from scepticism and from relativism, since it holds that a claim can be well supported, carefully tested and worth teaching while remaining open to revision. What is denied is finality, and what is retained is warrant.

[2] The word has acquired a second career. In recent work on artificial intelligence, a corrigible system is one disposed to accept correction, modification or shutdown from the operators who built it. The sense intended here is older and Popperian: a claim is corrigible when the world is permitted to overturn it. The two senses invert one another, and the inversion is instructive. A language model may be made corrigible to us while remaining incapable of corrigibility to the world, since nothing in its training obliges a claim to survive contact with an experiment. Our theories face the opposite requirement, being answerable to a world that neither built them nor can be persuaded to spare them.

tion or rate a symbol represents, and they survive intact when recall and algebra are both secure.

Such failures may be described as **ontological misclassifications**, and the corresponding educational capacity will be called **ontological fluency**: the ability to recognise what kind of quantity or relation a physics term denotes, how that status depends on the model being used, and which mathematical operations are licensed by it. A companion paper, *Ontological Fluency in Physics: Diagnostic and Classroom Tasks*, develops the diagnostic questions and classroom activities proposed here; the present essay establishes the conceptual argument.

The ontology–operation gap

Consider $F = ma$, read as "force is mass times acceleration". The sentence converts three symbols into three familiar nouns while settling almost nothing, since it does not say whether force is an object, a property, a cause, an interaction or the result of a multiplication, nor whether the equality reports a cause, a recipe or a joint constraint within a model.

A further difficulty is that the relation does definitional work while appearing to state a fact about the world, force and inertial mass being largely characterised through each other within the Newtonian scheme; a companion essay pursues that hinge structure, and it is noted here for contrast with an equation of state such as

$$\frac{PV}{T} = \text{constant},$$

which reports how three independently measurable properties of a fixed sample are found to covary, and which can fail, as real gases oblige it to. The two license different operations, and a student who reads both as formulae to rearrange has flattened a distinction that matters more than the algebra does. The same holds within Newton's law itself, whose explicit forms, $\sum \mathbf{F} = m\mathbf{a}$ and $\sum_i \mathbf{F}_i = d\mathbf{p}/dt$, each clarify something while adding to the visible symbolic burden, so that the compact version moves difficulty off the page rather than removing it.

"The simpler equation may remove notation by transferring its work into interpretation."

This is the **ontology–operation gap**: students are asked to operate on symbols before they have securely classified what the symbols represent.

From things to processes

Chi, Slotta and de Leeuw proposed that some scientific misconceptions persist because a learner assigns a scientific concept to an inappropriate ontological category [1], treating a process as though it were matter, or a constraint-based interaction as though it were a substance moving from place to place. Reiner, Slotta, Chi and Resnick developed this through examples of naive physics reasoning in which heat, light, force and electric current are treated in substance-like ways [2]; such reasoning can be internally coherent even when scientifically wrong. If current is imagined as material substance, it is reasonable to expect some of it to be consumed by a bulb; if force is imagined as an impetus placed inside a moving body, it is reasonable to expect the body to stop once the impetus is used up; if heat is imagined as a fluid held by an object, it is reasonable to speak of heat being stored or lost.

This changes the diagnostic question from "which fact has the student forgotten?" to "what must the student believe this quantity *is* for the answer to make sense?"

Nominalisation and reification

Scientific language depends on nominalisation, the transformation of a verb or relation into a noun: a body accelerates, and we speak of its acceleration; energy transfers, and we speak of an energy transfer; charges interact, and we speak of an electric field or force. Nominalisation is indispensable, since it lets processes become objects of thought that can be compared, quantified and related to other concepts. The danger begins when the noun is reified, when an abstraction is treated as a concrete agent: "the body accelerates" becomes "the body has acceleration" becomes "acceleration acts on the body"; "the velocity remains constant" becomes "the body has inertia" becomes "inertia keeps it moving".

Wittgenstein's later philosophy is useful here, since it directs attention towards how a word functions within a practice rather than towards an imagined essence behind it [8]. "The object continues because of inertia" sounds explanatory, yet if "inertia" means only the tendency expressed by Newton's first law, the sentence risks explaining continued motion by renaming continued motion: it has transformed a pattern into a noun and then assigned the noun causal power.

From sentence grammar to physical ontology

Before students can interpret the ontology of an equation, they often need to recover the relational structure of the sentence it is built from. A useful prompt is *who or what did what to whom*, a simplified subject–predicate–object analysis. Consider: "A slow-moving neutron collides with a heavy, unstable uranium nucleus, causing it to split into two lighter daughter nuclei and release further neutrons." This parses as neutron → collides with → uranium nucleus, with the consequence, collision → results in → fission and neutron release, represented separately. Such parsing can aid comprehension even before the physics is understood.

Grammatical clarity can still conceal physical ambiguity. "Gravity accelerates water down steep gradients" has a clean subject–predicate–object form, gravity → accelerates → water. The grammatical subject also makes "gravity" sound like an agent performing an action. A more explicit account restores the interacting systems: Earth → exerts gravitational force on → water, and separately, resultant force on the water → is related to → its acceleration. Syntax tells us how a sentence is organised; ontological analysis asks whether that organisation represents the physics appropriately. Abstractions, laws and nominalised processes routinely occupy the grammatical subject position, as in "gravity pulls", "inertia keeps it moving", "heat seeks equilibrium", "entropy drives the process", and each is well formed while still risking a misleading ontology.

The passive voice and the deletion of agency

A complementary manoeuvre works in the opposite direction. Where the agent error installs a false actor in the subject position, the passive voice removes the actor altogether, and scientific prose uses it constantly and often correctly. “The solution was heated to 60°C” suppresses the experimenter because the identity of the experimenter is irrelevant to the result, which is the passive doing honest work in the service of reproducibility. “A force is exerted on the block” suppresses something rather less dispensable, since the second interacting system has quietly vanished and the sentence now reads as though force arrived from nowhere and settled on the block.

Students already own the analytic tool for this, and they own it from elsewhere in the school. English and media-studies teaching devotes considerable attention to agency deletion, to “mistakes were made”, to “jobs will be lost”, to the passive constructions of official apology and press release in which the responsible party is edited out of a sentence that remains perfectly grammatical. The comprehension question put to students there, who is missing from this sentence and why, is exactly the question physics needs. Naming that continuity is worth a minute of lesson time, because a class that will happily interrogate a politician’s syntax will accept “heat is lost” from a textbook without noticing that it has been told nothing about which system lost what to which other system.

The physics test is therefore whether the deleted agent was a genuine second system or an irrelevant human. “The wire was connected to the cell” loses nobody who matters. “Energy is lost as heat” loses the destination, and restoring it, energy is transferred from the filament to the surrounding air, recovers a boundary and a direction that the passive had folded away. “The object is acted upon by gravity” preserves the agent grammatically while still leaving “gravity” standing in for the Earth, which is the two errors combined, an abstraction promoted to actor inside a construction that was supposed to be scrupulous about naming one.

Subject–predicate–object work should therefore proceed through three questions: what is the grammatical subject, predicate and object; what physical entities or relations do those words represent; and has the grammar turned a process or relation into a thing or agent? Students first parse the sentence, then repair its physical ontology, system → interaction or relation → second system. Knowledge graphs make this repair visible: a weak triple such as object → has → force can be replaced by rope → exerts force on → object, exposing the second system and the directed relation that the flat “has” concealed. The same procedure clarifies causal chains, as when the fission sentence above is expanded into three linked triples rather than treated as one undifferentiated event, and it prepares students to see which part of a chain is observed, which is inferred, and which is supplied by a model.

The pedagogical sequence is syntax → relation → ontology → equation, extending the earlier chain phenomenon → sentence → system → entities → relations → quantities → equation → operation. This matters because many problems are first encountered as prose: a student can fail because the syntax was never parsed, even when the relevant equation is known, or can parse the syntax correctly and still accept a misleading ontology. Syntactic fluency identifies who or what is grammatically doing something; ontological fluency asks whether the supposed actor is an object, interaction, process, field, rate, ratio or model construct. Physics problem solving requires both.

Five forms of ontological misclassification

The misconceptions considered here fall into five overlapping families, useful for diagnosis even where the boundaries blur.

1. The substance error

A process, rate or bookkeeping quantity is treated as material stuff: “heat is stored in an object”, “current is used up by the bulb”, “energy is contained in the battery”. This language is often productive, supporting diagrams and conservation reasoning, while also inviting an underlying-substance picture. Current, $I = \Delta Q/\Delta t$, is a rate at which charge crosses a boundary rather than a quantity of charge carried inside a component; in a steady series circuit the same rate occurs at every point, though energy is transferred at each component. A student predicting a lower current after the bulb than before may be preserving a coherent substance ontology in which some current was consumed, so correcting “current is the same everywhere” can change the answer without touching the underlying image, which then reappears in another context.

2. The agent error

A theoretical term is treated as an actor that makes events happen: “gravity pulls the object down”, “the battery pushes electrons around the circuit”, “inertia keeps the object moving”. These phrases compress difficult relations into accessible verbs, which is genuinely useful pedagogically. Their usefulness fails when metaphorical agency is mistaken for mechanism. “Gravity pulls” works adequately in a near-Earth Newtonian model but becomes unstable once a student must distinguish gravitational force from field strength, or compare inertial and gravitational mass. The agent error is especially prone to generating spurious forces, such as a forward “force of motion” drawn on a coasting object, where motion is treated as evidence of a sustaining agent.

3. The attribute error

A relational quantity is treated as an intrinsic possession of a single object: “the object has a force of 20 N”, “the wire has a current of 3 A”. These sentences share grammar while expressing different physical relations. Mass can often be treated as a property of the chosen system. A force is force *on* one system *by* another, while current is defined through charge crossing a section rather than possessed by an isolated wire in the way the wire possesses a length. Replacing object → has → force with Earth → exerts gravitational force on → object, or wire → has → current with charge → crosses section per unit time → current, restores the interaction or rate that the flattened predicate “has” removed. Knowledge-graph edges disclose exactly these ontological assumptions: whether a quantity is a property, an interaction, a measurement or a theoretical construct.

4. The process–state error

A state-like quantity is confused with a rate, change or transfer, as in the pairs velocity/acceleration, energy/power, charge/current, and temperature/heat transfer, with equations of the common form $a = \Delta v/\Delta t$, $P = \Delta E/\Delta t$, $I = \Delta Q/\Delta t$. The shared mathematical form is a quotient, while the physical meanings differ, and a student may manipulate the formula correctly while imagining power as an amount of energy or current as accumulated charge. School algebra presents the fraction as an operation to perform; physics requires the quotient to be read simultaneously as division and as a rate of transfer per unit time.

5. The model–object error

A representation is mistaken for the phenomenon it represents: magnetic field lines are treated as though they emerge physically from a pole, a ray as the literal route of a photon, an ideal gas as an accurate picture of real molecules. These drawings and models make reasoning possible. Their usefulness depends on a disciplined distinction between model and world. The phrase "real gases deviate from ideal behaviour" is revealing, since it grammatically treats the real gas as a poorly behaved imitation of the model, when in fact it is the model that ceases to approximate the gas adequately under certain conditions; the direction of comparison has been reversed.

Intensive and extensive: a distinction the classroom rarely names

Chemistry teaching draws a line that physics teaching often leaves implicit. An **extensive** quantity scales with the amount of material present, so that doubling the sample doubles it; mass, volume, charge, internal energy and heat capacity behave this way. An **intensive** quantity does not, and takes the same value for a sample and for any part of it; temperature, pressure, density, resistivity, specific heat capacity and refractive index behave this way. Many intensive quantities are ratios of two extensive ones, which is why they so often arrive as a fraction that students are told to evaluate.

Density, $\rho = m/V$, is the workhorse example. The arithmetic is elementary, and students commonly meet density described as "the amount of stuff packed into a given volume", a memorable phrase in which "stuff" is ontologically vague and "packed" suggests particles compressed into a container, so that density comes to seem like the amount itself. A student who says that a larger object must be denser has confused an extensive quantity with an intensive one; a student who inserts the volume of one body and the mass of another has failed to hold the sample fixed across both measurements; a student who treats ρ as an additional substance inside the object is reading the equation as a recipe in which mass and volume produce a third ingredient. Each error is a misclassification, and none is arithmetical.

The distinction does real work as soon as it is named. Hydrostatic pressure,

$$P = \rho g h,$$

is a good place to see it, because the equation contains no extensive quantity at all and the results students find counterintuitive follow directly from that. Pressure at depth depends on the density of the liquid, the field strength and the depth, and not on the width of the vessel, the shape of the column or the total mass of water above; a narrow tube and a wide reservoir filled to the same height press equally on their bases. Students who expect more water to mean more pressure are importing an extensive intuition into an intensive relation. The same reading illuminates the gas law: P and T are intensive, V is extensive, and PV/T = constant holds for a fixed amount of gas precisely because the extensive quantity has been pinned down, which is what the molar form makes explicit.

A short classroom routine follows. Given any quantity, ask what happens to it if the sample is cut in half. Mass halves, volume halves, density does not; charge halves, potential difference does not; heat capacity halves, specific heat capacity does not. The test is quick, it is performed on the physics rather than on the symbols, and it exposes exactly the category confusion that rearranging the formula would conceal.

Reversal errors and noun-label algebra

Clement's classic reversal-error study showed that even advanced students can struggle to translate verbal relationships into algebra [3]. Given "there are six times as many students as professors", students often write $6S = P$ instead of $S = 6P$, a word-order matching error compounded by a static-comparison reading in which the coefficient attaches to the larger group; the letters function as compressed noun labels, $S \equiv$students, rather than as variables representing counts.

Physics equations invite the same mistake. F, m and a may be read simply as the words "force", "mass" and "acceleration", so that $F = ma$ resembles a sentence assembled from labelled nouns. A physics symbol is an abbreviation for a whole cluster of commitments, since it represents a quantity belonging to a system, carrying dimensions and units, embedded in a model, and often possessing direction. Reading $F = ma$ as "force equals mass times acceleration" is only the beginning of interpretation; it does not establish which force, whose mass, which component of acceleration, or under what conditions the relation applies.

Equations as symbolic forms

Sherin's account of **symbolic forms** bridges mathematical structure and conceptual meaning [4]: experts recognise recurring patterns bound to intuitive schemas rather than decoding each symbol independently, seeing in PV/T = constant a constraint linking properties of a fixed sample, and in $V = IR$ a proportionality involving a system parameter. The equation has become a conceptual object, which is why notation that looks complicated to a novice can look economical to an expert who has chunked form and meaning together. Redish describes mathematics in physics as a disciplinary dialect blending symbolism with physical interpretation [5], Kuo and colleagues find conceptual and formal reasoning blended opportunistically throughout manipulation rather than only around it [6], and Torigoe notes that a symbolic equation represents a class of systems, its apparent indeterminacy being part of its power [7]. Students must therefore learn how an equation can be read, unpacked, specialised and reconstructed, which is a larger task than recalling it.

From phenomenon to operation

A physics problem can be represented as phenomenon → system → entities → relations → quantities → equation → operation. Much routine instruction begins near the end, at quantities → equation → operation: values are extracted, a formula selected, rearranged and substituted. This can succeed when a problem closely resembles a learned example. Transfer stays fragile while the earlier modelling decisions remain implicit. Before applying Newton's second law, the solver must decide what system is being analysed, which interactions cross its boundary, which quantities are vectors, what coordinate convention applies, whether the frame is inertial, whether the mass is constant, and whether the acceleration concerns speed, direction or both. These are ontological and representational decisions that determine what the algebra means; a formula sheet can remove the need to recall the final equation. The prior acts of system selection and model construction remain.

Misconceptions as alternative ontologies

Some errors are better understood as consequences of a coherent alternative ontology than as defective propositions awaiting replacement. If force is a motive property possessed by a body, then motion is evidence that force remains inside it, so a moving object seems to require a forward force; the scientific account instead distinguishes velocity from acceleration and force from motion, so that $\sum \mathbf{F} = \mathbf{0}$ implies $\mathbf{a} = \mathbf{0}$ while $\mathbf{v}$ may remain constant and non-zero, which requires reclassifying force from a stored property of motion to an interaction contributing to change of momentum. If force alone is treated as the determinant of acceleration, a heavier body seems to fall faster because $F_g = mg$ is larger for it; Newton's second law supplies the inertial response, $a = F_g/m = mg/m = g$, an algebraic cancellation whose meaning is ontological, since the same mass that increases gravitational force also measures inertial response. If current is a substance, transferring energy seems to require consuming the carrier, so a bulb appears to use up current; the relevant distinction is between $I = \Delta Q/\Delta t$ as a charge-transfer rate and $P = VI$ as an energy-transfer rate, the bulb transferring energy without consuming steady current. And $P = E/t$ can be manipulated correctly while power is still treated as an amount, requiring the student to reclassify power as a rate. In each case the wrong answer is rational within the student's category system, and correction requires a shift in what the quantity is taken to be.

Ontological fluency

Ontological fluency is the capacity to identify whether a physics term is functioning as an object, property, relation, interaction, state, process, rate, field, model parameter or representational construct, and to use that classification appropriately when interpreting and manipulating equations. This does not require students to settle a final metaphysical position on whether fields or forces are "really real"; the classroom aim is operational; students should recognise that force is relational and vectorial, that acceleration is a rate relative to a frame, that current and power are rates, that intensive and extensive quantities behave differently under division of the sample, that energy is defined relative to a chosen system, that a field assigns quantities to locations, that a field line is a representation rather than a physical filament, and that an ideal gas is a model system rather than a perfect real gas.

Ontological fluency should be distinguished from **symbolic fluency**, the ability to read and rearrange expressions, and from **model-selection fluency**, the ability to decide which idealisation or equation applies to a situation; physics problem-solving fluency is roughly the sum of all three. Algebraic competence alone does not constitute physical problem-solving competence.

The classroom programme

The companion paper develops a classroom programme from this framework, grouped into five task families: classifying a term (mass, force, current, field) using categories such as property, relation, rate, interaction and model parameter; repairing sentences whose grammar conceals a physical relation, such as turning "the object has a force of 20 N" into "the rope exerts a 20 N force on the object"; expanding an equation into a subject–predicate–object graph, linking free-body diagrams to equation construction; diagnosing an incorrect solution by asking what the solver must believe the quantity *is* for their answer to make sense; and moving between representations, phenomenon, sentence, diagram, graph, equation and units, since each reveals some commitments while suppressing others. The full tasks, teacher notes and diagnostic responses are published in the companion document:

Ontological Fluency in Physics: Diagnostic and Classroom Tasks.

The Ontology Atlas, an interactive map of the same relations.

Ontology and cognitive load

This framework sharpens an earlier criticism of cognitive load as educational phlogiston. Cognitive load is a genuine constraint, since a learner cannot indefinitely coordinate separate symbols, definitions, diagrams and units. Saying that $I = \Delta Q/\Delta t$ is difficult because it imposes excessive load still leaves open why the elements remain separate. For the expert the expression may function as one conceptual unit, a rate at which charge crosses a boundary; for the novice it may remain two symbols, a division sign, an unfamiliar difference operator and several possible readings. Ontological fluency helps explain what has been chunked and identifies the category structure that lets several symbols cohere into one meaningful form. Cognitive load describes the burden of coordinating the elements; ontological fluency helps explain whether those elements can cohere at all, and without that second analysis, cognitive load risks becoming a label attached after meaning has already failed.

Teaching truths, tools and disciplined fictions

Nothing here requires scientific terms to be treated as fictions. A realist may take fields and forces to refer to genuine structures, a model-based philosopher may emphasise their domain-bound function, and a Wittgensteinian may attend to the practices that fix their meaning; the educational problem arises before any of these positions needs settling, since what students require is knowledge of how a concept functions within the model in use. However a magnetic field is understood philosophically, field lines must not be confused with material threads, and however force is understood, it must not be confused with velocity or stored motion. The aim is a disciplined pluralism in which students use a representation without mistaking it for the phenomenon, a metaphor without mistaking it for a mechanism, and a noun without assuming it names a substance.

Idealisation and the student's misgiving

Cartwright argues that the fundamental laws achieve their generality and explanatory reach at a cost, since what they describe with precision are the highly idealised models we construct rather than the untidy situations we encounter [9]. The equations hold exactly of frictionless planes, isolated systems, point masses, ideal gases and infinite wires, and hold of the world by way of those constructions, within domains where the idealisation is good enough for the purpose at hand. There is a family resemblance to effective field theory, which retains the degrees of freedom that matter at the scale under consideration while absorbing the rest into parameters and corrections, and which is expected to yield to another description when the scale or the required precision changes. School modelling is far simpler, though the governing discipline is the same: the modeller asks what may be left out, what follows from leaving it out, and what observation would show that the permission has expired.

This judgement is where the modeller's skill becomes visible, and it is routinely performed offstage. Students meet the completed model, with the world already partitioned into system and surroundings, some interactions retained and others suppressed, quantities defined and a tolerance assumed, so that the equation appears to follow directly from the phenomenon and the student's task is described as applying it. Much of what distinguishes physical reasoning from algebra lies in reconstructing exactly those prior decisions.

Students feel the omission before they can name it, and their complaints are the evidence. The pupil who objects that a real ball would not keep rolling, that a real circuit gets warm, or that a real gas cannot be made of points with no volume has noticed the gap the model was built to open, and is registering, in the only vocabulary available, that the syntax of the equation and the furniture of the world have come apart. Much teaching clears such objections away so the calculation can proceed, defensibly enough given forty minutes and an examination that will ask for the idealised answer, at the cost of training students to suppress the disposition that makes a physicist. A small change preserves it: name the idealisation, say what it discarded, and say what would happen to the answer if that were restored.

Ontological fluency therefore includes the capacity to inhabit a constructed world without forgetting that it was constructed, treating its entities seriously enough to reason with and provisionally enough to see where they cease to serve. To ask what kind of thing a symbol denotes is to ask which constructed system it belongs to, and that is to ask how far the construction can be trusted before the world declines to cooperate. The student's misgiving, taken seriously, is a small rehearsal of the tumbler.

Conclusion: the nouns beneath the numbers

Physics equations are often presented as the point at which language has become precise, yet their symbols inherit the ambiguities of the words beneath them. The letter F can conceal an interaction, a a changing vector, I a rate across a boundary, ρ a ratio property, and E a bookkeeping quantity defined relative to a system. Algebra compresses these ontological questions; it does not settle them.

This is why students can remember a formula, substitute values correctly in familiar exercises, and still fail when the surface context changes. They have learned an operation without the conceptual classification that gives it physical meaning. The resulting errors are often coherent consequences of treating rates as substances, interactions as possessions, models as objects, or abstract nouns as agents.

Physics problem solving begins before the equation is selected, with the construction of a world in which the equation can mean something: what the system is, which entities are present, which relations connect them, which quantities represent those relations, and which operations those quantities permit; symbolic manipulation then becomes physical reasoning. Ontological fluency names this neglected capacity, offering a route from philosophical reflection to classroom diagnosis, and from the grammar of scientific nouns to the practical errors found in student calculations.

The same capacity also carries a wider significance. Steve Jobs described a rock tumbler, a coffee can filled with ordinary stones and a little grit, turning noisily overnight until the stones emerged polished. Physics places its conceptual objects in a comparable tumbler. Observation, measurement and intervention grind against the claims, retaining what holds and reshaping what does not.

An equation becomes physical when its symbols are made answerable to a system, a measurement and a possible refusal. Ontological fluency is the capacity to make that commitment consciously: to know what has been introduced, what work it performs, and what could require its revision. Our desire to know may remain incorrigible; the objects through which we know must remain corrigible.

A machine may rearrange the tokens, whereas physics begins where the tokens are put into the world and the world is allowed to push back.

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A companion task paper develops the diagnostic questions, sentence repairs, knowledge graphs and equation-unpacking activities outlined here.